MODULE 04 · DOMAIN 1 - SOLUTION DESIGN AND ARCHITECTURE

Multi-Agent Systems and Orchestration

When a team of agents is worth the tokens, how to delegate well, and where the agent loop should run.

DOMAIN 1 of 7 EXAM WEIGHT 17% MODULE 4 of 38

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A team of agents is not smarter by default.
It is **parallel, isolated and expensive**.

Orchestrator and workers, as Anthropic built it

Subagents
Three to five in parallel
Own context each

Tools
Search and fetch
Guidance per subagent

Lead agent
Reads the request
Decides the subtasks
Plans, delegates, combines

Memory
Plan saved outside
Survives long runs

Citation step
Links claims to sources
Runs at the end

THE TAKEAWAY

The lead never does the digging itself. It plans, delegates and synthesises.

What Anthropic measured (June 2025)

FINDING	FIGURE
Multi-agent over single agent, research eval	+90.2%
Agent tokens compared with chat	about 4x
Multi-agent tokens compared with chat	about 15x
Performance explained by token use	80%
Research time cut by parallelism	up to 90%

ON THE EXAM

Most of the gain is more tokens, spent in parallel. Worth it only when the task is valuable.

When a team of agents fits

FITS

Broad and parallel

- Many independent directions
- More material than one context holds
- High-value result
- Many tools and sources

FAILS

Shared and coupled

- Every agent needs the same context
- Tight dependencies between steps
- Most coding changes
- Cheap or real-time requests

THE TIEBREAKER

Independent directions favour a team. Shared context and tight dependencies favour one agent.

What every subagent must be told

- **An objective:** what this subagent is finding out, and why.
- **An output format:** what to hand back, and how long.
- **Tools and sources:** which to use, and which to avoid.
- **Boundaries:** where its task stops and another's begins.
- **Effort to match:** a simple lookup is one agent and a few calls.

> ON THE EXAM
>
> Anthropic's early lead agent wrote one-line briefs. Subagents duplicated each other's work.

The quiet benefit: context isolation

ONE AGENT DOES EVERYTHING

A crowded context

- Raw tool output piles up
- Early findings get buried
- Accuracy falls as it fills

SUBAGENTS EXPLORE SEPARATELY

A clean lead context

- Each digs in a fresh window
- Returns a short, distilled summary
- The lead keeps only what matters

THE TIEBREAKER

Subagents are a context-management tool as much as a speed tool.

Splitting one overloaded agent

- **Symptoms:** tools across unrelated domains, overlapping descriptions, falling accuracy.
- **Split by domain first:** one focused agent per domain, with its own tools and prompt.
- **Then route** requests to the right domain agent.
- **A router in front of the same bloated agent** hides the problem rather than fixing it.

> ON THE EXAM
>
> Decompose the capability first. A router only helps once there is something clean to route to.

Where the agent loop runs

OPTION	WHAT YOU GET
Messages API	You write the loop and run everything; full control
Agent SDK	Claude Code's loop and tools, as a library, in your process
Managed Agents	Anthropic runs the loop, sandbox, sessions and credentials

CURRENCY CHECK

Managed Agents has been in public beta since 8 April 2026. The Agent SDK was formerly the Claude Code SDK.

Choosing the runtime: constraints first

- **Compliance:** Managed Agents is not currently eligible for ZDR or HIPAA BAA coverage.
- **Platform:** Managed Agents runs on the Claude API and Claude Platform on AWS only.
- **Data:** sessions and history are stored by Anthropic, even with a self-hosted sandbox.
- **Then weigh savings:** long-running, asynchronous work is where hosting pays most.

> ON THE EXAM
>
> Eliminate on compliance, platform and data location. Only then compare operational savings.

An investment research firm produces a weekly report on 40 listed companies. Each needs its own investigation of filings, news and analyst commentary, followed by a sector-wide synthesis.

A single agent runs for hours, fills its context, and loses track of companies it researched early.

What should the architect design?

A. A lead agent that assigns groups of companies to parallel subagents, each with a clear brief, then synthesises

B. The same single agent on the largest context window at the highest effort

C. A prompt chain that researches the companies one after another

D. Forty subagents, each told to research the market and report back

◆ PAUSE AND DECIDE

What should the architect design?

A A lead agent that assigns groups of companies to parallel subagents, each with a clear brief, then synthesises

Broad, independent, valuable work: parallel subagents with clean contexts and real briefs.

B The same single agent on the largest context window at the highest effort

A bigger window still fills, and recall still falls as it does.

C A prompt chain that researches the companies one after another

Still slow, and still one context accumulating forty companies.

D Forty subagents, each told to research the market and report back

One vague brief for all: duplicated work and nothing to combine cleanly.

THE SIGNAL

Many independent directions, too much for one context, a valuable result: orchestrator and subagents.

> Tallyhawk plans to refactor its billing module. The changes are tightly coupled: each depends on decisions made in the one before, and all share the same data model.
>
> An engineer proposes five agents working in parallel, each owning one part of the module.

What should Corvane advise?

A. One agent makes the changes in sequence, using a subagent only for read-only exploration

B. Five parallel agents, each owning one part, merging at the end

C. An orchestrator that splits the work by file and assigns workers

D. Three agents implement the whole refactor independently, and the best result is kept

◆ PAUSE AND DECIDE

What should Corvane advise?

A One agent makes the changes in sequence, using a subagent only for read-only exploration

Coupled steps need one shared context. Exploration can still be offloaded.

B Five parallel agents, each owning one part, merging at the end

Each part depends on the others. Parallel edits will conflict.

C An orchestrator that splits the work by file and assigns workers

Still parallel edits to coupled code, with a merge problem at the end.

D Three agents implement the whole refactor independently, and the best result is kept

Triples the cost for a task one careful agent can do.

THE SIGNAL

Tight dependencies and a shared data model: one agent, not a team.

> A market-analysis agent spawns subagents to research competitors. Their reports overlap heavily, two cover the same company, and one investigated an unrelated industry.
>
> The logs show the lead agent's briefs are single lines, such as "look into competitors".

What is the most direct fix?

A Spawn more subagents so that coverage improves

B Have the lead give each subagent an objective, an output format, tool and source guidance, and clear boundaries

C Move every subagent to the most capable model tier

D Let all subagents share one conversation history so they can see each other's work

◆ PAUSE AND DECIDE

What is the most direct fix?

A Spawn more subagents so that coverage improves

More agents with vague briefs means more duplication.

B Have the lead give each subagent an objective, an output format, tool and source guidance, and clear boundaries

Fixes the cause the logs show: vague delegation.

C Move every subagent to the most capable model tier

A stronger model still cannot know a boundary it was never given.

D Let all subagents share one conversation history so they can see each other's work

Throws away context isolation, the main benefit of subagents.

THE SIGNAL

Overlap and drift with one-line briefs: fix the delegation, not the model or the headcount.

> Quillon's discharge-summary agent reads lab results, nursing notes, medication charts and consultant letters. Its context fills with raw text, and summaries begin to omit findings it read early in the run.
>
> Each record area can be read independently, and the summary must cite where each finding came from.

Which change best addresses the problem?

A. Instruct the agent to remember every finding carefully

B. Automatically clear older tool results whenever the context fills

C. Subagents read each record area in their own context and return condensed, cited findings to the lead

D. Add a second agent that reads the whole record again to check the summary

◆ PAUSE AND DECIDE

Which change best addresses the problem?

A Instruct the agent to remember every finding carefully

An instruction cannot stop recall falling as the context fills.

B Automatically clear older tool results whenever the context fills

Frees space by discarding exactly the early findings being lost.

C Subagents read each record area in their own context and return condensed, cited findings to the lead

Isolates the raw text; the lead keeps only distilled, cited findings.

D Add a second agent that reads the whole record again to check the summary

The checker's context fills in exactly the same way.

THE SIGNAL

A crowded context with separable inputs: isolate the reading in subagents, keep the lead clean.

> Mereside wants long-running agents to process mortgage document packs overnight. Its policy requires processing in the UK or EU, which is why it uses Claude on Amazon Bedrock in a European region.
>
> A developer suggests Claude Managed Agents, to avoid running sandboxes and sessions in-house.

What should Corvane recommend?

A. Managed Agents with Anthropic's cloud sandbox

B. Managed Agents with a self-hosted sandbox in Mereside's own data centre

C. Move to the first-party API with US inference so Managed Agents can be used

D. The Agent SDK running in Mereside's environment, calling Claude on Bedrock in its European region

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Managed Agents with Anthropic's cloud sandbox
Not offered on Bedrock, so it leaves the European processing route.

B. Managed Agents with a self-hosted sandbox in Mereside's own data centre
Tools run locally, but the sessions themselves still run in Anthropic's service.

C. Move to the first-party API with US inference so Managed Agents can be used
Breaks the residency policy outright to gain convenience.

D. The Agent SDK running in Mereside's environment, calling Claude on Bedrock in its European region
A full agent loop that stays on the compliant platform.

THE SIGNAL

A data-location constraint the hosted runtime cannot meet: run the loop yourself on the compliant platform.

> A team is moving parts of a long research task from one agent to subagents coordinated by a lead.
>
> The product owner asks what the change will actually buy them.

Which two are genuine benefits of using subagents here? Choose two.

A Subagents automatically share the lead's full conversation history

B Total token use falls compared with a single agent

C Each subagent explores in a clean context, so the lead's context stays focused

D Independent subtasks run in parallel, cutting elapsed time

E Results become deterministic and repeatable run to run

◆ PAUSE AND DECIDE

Which two are genuine benefits of using subagents here? Choose two.

A Subagents automatically share the lead's full conversation history

False. A subagent sees only what it is given.

B Total token use falls compared with a single agent

False. Multi-agent runs use far more tokens.

C Each subagent explores in a clean context, so the lead's context stays focused

True: context isolation.

D Independent subtasks run in parallel, cutting elapsed time

True: parallelism.

E Results become deterministic and repeatable run to run

False. More model calls add variation, not remove it.

THE SIGNAL

Subagents buy two things: clean contexts and parallel time. They cost tokens.

In five lines

- **A team of agents buys parallelism and clean contexts**, at many times the tokens.
- **Fits broad, independent, valuable work.** Fails on shared context and tight coupling.
- **Brief every subagent:** objective, output format, tools and sources, boundaries.
- **Split an overloaded agent by domain** before putting a router in front of it.
- **Pick the runtime constraints first:** Managed Agents is not ZDR, HIPAA or Bedrock.

COMING UP

Next: the end-to-end architecture around the model. Input, processing, output and feedback.